

Hierarchical Leverage for Crisis-Resilient Portfolios

Lessons from Bitcoin's October 2025 Liquidity Crash

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Context & Motivation

- Crypto markets combine **high leverage**, **thin liquidity**, and rapid **deleveraging**.
- Oct 2025: Bitcoin fell $\sim 8\%$ with record liquidations across derivatives venues.
- Flat (linear) leverage scales losses nearly linearly with expected leverage $E[L]$.
- **Goal:** Design a **Hierarchical Leverage Framework (HLF)** that compresses tail losses while preserving scalable exposure.

Universe, Layers, Constraints

Leverage layers. $\mathcal{L} = \{L_1, \dots, L_N\}$ with $1 = L_1 < \dots < L_N$ (e.g., 1, 2, 3, 5, 7, 10, 15, 20).

Weights. $w_i \geq 0$ with

$$\sum_{i=1}^N w_i = 1, \quad E[L] \equiv \sum_{i=1}^N w_i L_i.$$

Shock loss (first-order). For return shock $\Delta \approx -\alpha$,

$$\text{Portfolio loss} \approx \sum_{i=1}^N w_i L_i \Delta = E[L] \cdot \Delta.$$

Design idea: keep $E[L]$ moderate while distributing risk nonlinearly across layers (tiny w at high L).

Option A: Exponential Decay (Risk-first)

Definition ($k > 0$).

$$\tilde{w}_i = \exp(-k(L_i - L_{\min})), \quad w_i = \frac{\tilde{w}_i}{\sum_{j=1}^N \tilde{w}_j}.$$

Properties.

- Monotone: higher $L \Rightarrow$ exponentially smaller w .
- Single “risk-temperature” k controls tail sharpness ($k \uparrow \Rightarrow$ safer).
- $E[L]$ is implied by k (or solve k by bisection to hit a target $E[L]$).

Option B: Power-Law Decay (Scale-Aware)

Definition ($\beta > 0$).

$$\tilde{w}_i = L_i^{-\beta}, \quad w_i = \frac{\tilde{w}_i}{\sum_{j=1}^N \tilde{w}_j}.$$

Rationale & trade-offs.

- Reflects scale invariance; slower decay than exponential.
- Larger $\beta \Rightarrow$ safer (lower tail weight); smaller $\beta \Rightarrow$ hotter (higher $E[L]$).

Option C: Entropy-Constrained (MaxEnt) with Target Exposure

Program.

$$\begin{aligned} \max_{w \geq 0} H(w) &= - \sum_{i=1}^N w_i \ln w_i \\ \text{s.t. } \sum_{i=1}^N w_i &= 1, \quad \sum_{i=1}^N w_i L_i = L_{\text{target}}. \end{aligned}$$

KKT solution.

$$w_i^* = \frac{e^{-\lambda L_i}}{\sum_{j=1}^N e^{-\lambda L_j}}, \quad \sum_{i=1}^N w_i^* L_i = L_{\text{target}} \text{ (solve for } \lambda \text{)}.$$

Notes. Strictly concave $\Rightarrow$ unique optimum; solve λ via bisection/Newton (stable).

Risk Budgeting by Layer

Let σ be realized vol, q_α a VaR/ES quantile. Layer i :

$$\text{VaR}_i \approx q_\alpha L_i w_i \sigma, \quad \sum_i \text{VaR}_i \leq \text{RiskBudget}.$$

Control knobs: increase k or β (steepen decay) or reduce L_{target} (MaxEnt) until budget holds; then renormalize w .

Dynamic Crash Governor (Volatility Regimes)

Regime map (example).

$$\text{Regime}(\sigma) = \begin{cases} \text{Normal}, & \sigma < \sigma_1 \\ \text{Elevated}, & \sigma_1 \leq \sigma < \sigma_2 \\ \text{Extreme}, & \sigma \geq \sigma_2 \end{cases}$$

Active layers.

$$\mathcal{L}_{\text{active}} = \begin{cases} \{1\times, \dots, 20\times\}, & \text{Normal} \\ \{1\times, \dots, 10\times\}, & \text{Elevated} \\ \{1\times, \dots, 5\times\}, & \text{Extreme} \end{cases}$$

Renormalize w on $\mathcal{L}_{\text{active}}$ (same formula).

Calibration (Illustrative)

- Layers: $L = \{1, 2, 3, 5, 7, 10, 15, 20\}$ (includes spot).
- Exponential $k=0.45$; Power-law $\beta=1.1$; MaxEnt target $L_{\text{target}} \in [2.2, 2.5]$.
- Shock scenario: $\Delta = -8\%$.

Loss vs Expected Leverage

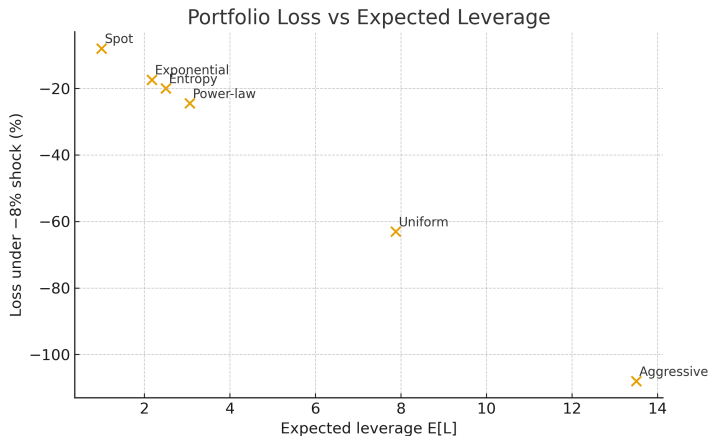
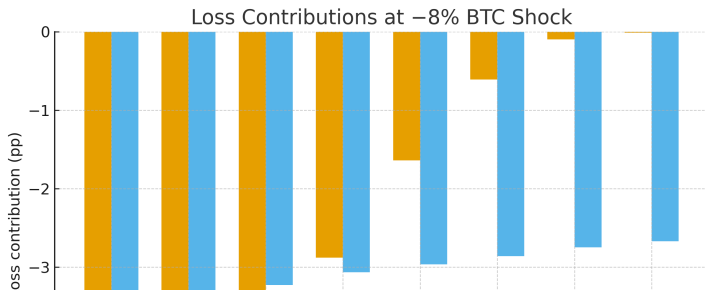
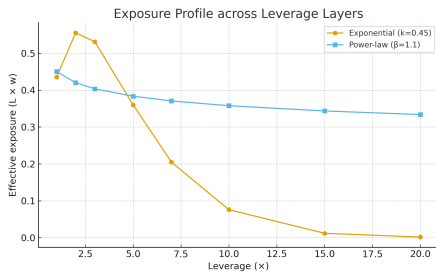
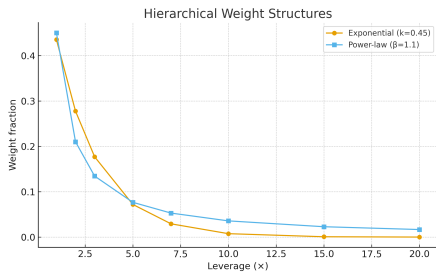


Figure: Hierarchical structures compress tail losses vs. flat leverage at similar $E[L]$.

Portfolio Loss Table at -8% Shock

Scheme	$E[L]$	Loss (%)	Comment
Spot-only	1.0	-8.0	Baseline
Uniform ($1-20\times$)	8.75	-70.0	Flat leverage
Aggressive ($10-20\times$)	13.5	-108.0	Liquidation
Exponential ($k=0.45$)	≈ 2.9	-23.2	Stable tail
Power-law ($\beta=1.1$)	≈ 3.3	-26.4	Hotter
MaxEnt ($L_{\text{target}}=2.5$)	2.5	-20.0	Best trade-off

Weight & Exposure Profiles



Hierarchical vs Flat (Visual)

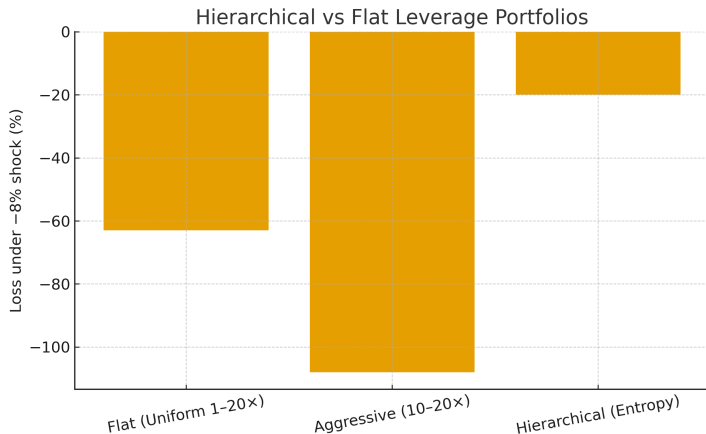


Figure: Hierarchical allocation absorbs shocks that destroy uniform portfolios.

Live Policy (Pseudo-code)

Input: L , scheme params (k / β / L_{target}),
 sigma thresholds (σ_1 , σ_2).

Loop each decision time:

- 1) Estimate realized volatility σ .
- 2) Set ActiveLayers per σ :
 - Normal $\rightarrow \{1x..20x\}$
 - Elevated $\rightarrow \{1x..10x\}$
 - Extreme $\rightarrow \{1x..5x\}$
- 3) Compute baseline weights w over ActiveLayers:
 - Exponential: $w_i \exp(-k (L_i - L_{\min}))$
 - Power-law : $w_i L_i^{(-\beta)}$
 - MaxEnt : $w_i \exp(-\lambda L_i)$ with $E[L]=L_{\text{target}}$
- 4) Risk budget (VaR/ES): if exceeded $\rightarrow$ increase k or β ,
 or decrease L_{target} ; then renormalize w .
- 5) Size trades: $\text{position}_i = w_i * \text{Capital} * L_i / \text{Price}$.

Conclusion

- Hierarchical leverage **compresses tail risk** vs. flat leverage at similar $E[L]$.
- Exponential & MaxEnt are robust defaults; Power-law provides controlled aggression.
- Crash-governor + risk budget keeps the system resilient in extremes (Oct 2025 BTC).
- Next: multi-asset (BTC, ETH, Gold), option overlays, agent-based stress tests.

Thank You

Questions?